## Problem

Sketch the magnitude Bode plot for

$$H(s)=\frac{10s(s+20)}{(s+1)(s^2+60s+400)}, \qquad s=j\omega$$

## Step-by-step solution

**Step 1** of 3

Consider the following transfer function:

$$\begin{aligned}H(s)&=\frac{10s(s+20)}{(s+1)(s^2+60s+400)}\\&=\frac{10s(s+20)}{(s+1)(s+7.6)(s+52.4)}\end{aligned}$$

Substitute $j\omega$ for $s$.

$$H(\omega)=\frac{10j\omega(j\omega+20)}{(j\omega+1)(j\omega+52.4)(j\omega+7.6)} \quad \ldots\ldots (1)$$

$$H(\omega)=\frac{(10)(20)j\omega\left(1+j\dfrac{\omega}{20}\right)}{(7.6)(52.4)(1+j\omega)\left(1+j\dfrac{\omega}{7.6}\right)\left(1+j\dfrac{\omega}{52.4}\right)}$$

$$=\frac{0.5j\omega\left(1+j\dfrac{\omega}{20}\right)}{(1+j\omega)\left(1+j\dfrac{\omega}{7.6}\right)\left(1+j\dfrac{\omega}{52.4}\right)}$$

The system has three poles at $-1,-52.4,-7.6$ and two zeroes, one at origin and one at $-20$

Determine the magnitude in dB of the constant.

$$20\log(0.5)=-6.021\text{ dB}$$

The zero at origin has a slope of $+20\text{ db/decade}$.

The pole at $1\text{ rad/s}$ has a slope of $-20\text{ db/decade}$.

The pole at $7.6\text{ rad/s}$ has a slope of $-20\text{ db/decade}$.

The zero at $20\text{ rad/s}$ has a slope of $+20\text{ db/decade}$.

The pole at $52.4\text{ rad/s}$ has a slope of $-20\text{ db/decade}$.

**Step 2** of 3

**Magnitude plot:**

The cut-off frequencies are,

$$\omega_{c1} = 1,$$
$$\omega_{c2} = 7.6,$$
$$\omega_{c3} = 20,$$
$$\omega_{c4} = 52.4$$

From equation (1), write the magnitude of $|H(\omega)|$.

$$|H(\omega)| = \frac{10(\omega)\left(\sqrt{(20)^2 + (\omega)^2}\right)}{\left(\sqrt{1+\omega^2}\right)\left(\sqrt{52.4^2+\omega^2}\right)\left(\sqrt{7.6^2+\omega^2}\right)}$$

Calculate the magnitude at $\omega = 1 \text{ rad/sec}$.

$$\begin{aligned}
|H(\omega)| &= 20\log\left[\frac{10(1)\left(\sqrt{(20)^2+(1)^2}\right)}{\left(\sqrt{1+(1)^2}\right)\left(\sqrt{52.4^2+(1)^2}\right)\left(\sqrt{7.6^2+(1)^2}\right)}\right] \\
&= 20\log\left[\frac{200.25}{(1.414)(52.41)(7.67)}\right] \\
&= 20\log(0.3523) \\
&= -9.056 \text{ db}
\end{aligned}$$

Calculate the magnitude at $\omega = 7.6 \text{ rad/sec}$

$$\begin{aligned}
|H(\omega)| &= 20\log\left[\frac{10(7.6)\left(\sqrt{(20)^2+(7.6)^2}\right)}{\left(\sqrt{1+(7.6)^2}\right)\left(\sqrt{52.4^2+(7.6)^2}\right)\left(\sqrt{7.6^2+(7.6)^2}\right)}\right] \\
&= 20\log\left[\frac{1626.04}{(7.67)(52.95)(10.75)}\right] \\
&= 20\log(0.3724) \\
&= -8.58 \text{ db}
\end{aligned}$$

Calculate the magnitude at $\omega = 20 \text{ rad/sec}$

$$\begin{aligned}
|H(\omega)| &= 20\log\left[\frac{10(20)\left(\sqrt{(20)^2+(20)^2}\right)}{\left(\sqrt{1+(20)^2}\right)\left(\sqrt{52.4^2+(20)^2}\right)\left(\sqrt{7.6^2+(20)^2}\right)}\right] \\
&= 20\log\left[\frac{5656.85}{(20.02)(56.09)(21.39)}\right] \\
&= 20\log(0.2355) \\
&= -12.56 \text{ db}
\end{aligned}$$

Calculate the magnitude at $\omega = 52.4 \text{ rad/sec}$

$$|H(\omega)| = 20\log\left[\frac{10(52.4)\left(\sqrt{(20)^2+(52.4)^2}\right)}{\left(\sqrt{1+(52.4)^2}\right)\left(\sqrt{52.4^2+(52.4)^2}\right)\left(\sqrt{7.6^2+(52.4)^2}\right)}\right]$$

$$= 20\log\left[\frac{29389.63}{(52.41)(74.105)(52.95)}\right]$$

$$= 20\log(0.1437)$$

$$= -16.85 \text{ db}$$

Sketch the asymptotic magnitude plot for the transfer function.

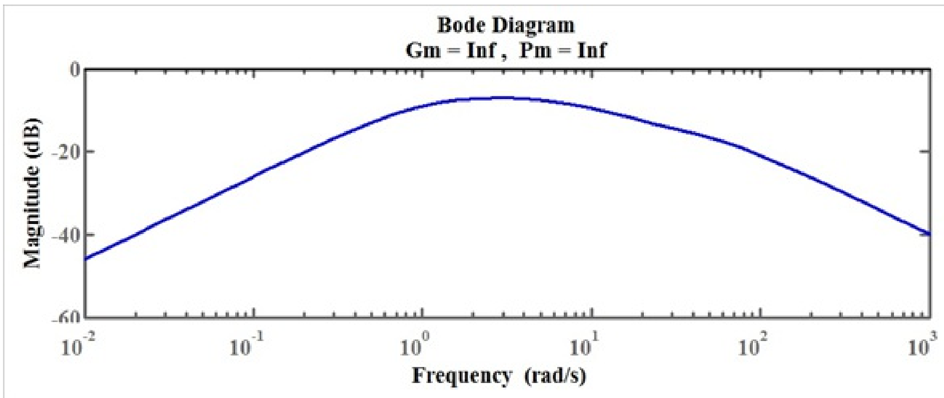


Figure 1

Thus, the Bode magnitude plot is sketched.